Delphic HSE Solutions Ltd

Toxicological Risk Assessment

Hand Sanitizer

PRODUCT DETAILS

Product Reference

Type of Product: Hand Sanitiser | Leave on

Physical Form: Liquid, Gel

Client: Zhejiang Ayan Biotech Co., Ltd
No.259 North Keyuan Rd, Technology &Industry Zone,
Ninghai, Ningbo,China

PHYSICAL / CHEMICAL CHARACTERISTICS

Appearance: Liquid
Odour: Fragranced
Melting Point: Not Provided
pH: Not Provided
Specific Gravity: Not Provided
Viscosity: Not Provided
Water Solubility: Not Provided
Log Kow: Not Provided
Particle Size: Not Provided

The physical-chemical data supplied for review suggests it would be unlikely to significantly contribute to the toxicological profile of the product under normal conditions of use.

FORMULATION OVERVIEW

The formulation below provides an overview of the composition, listing chemical name and quantities added. For further details please see Annex I - Raw Material Information.

Ingredients	CAS Number	[Product]
Ethyl Alcohol	64-17-5	67
Aqua	7732-18-5	32.029
Propylene Glycol	57-55-6; 4254-14-2	.5
Carbomer (NOS)	9007-16-3 ; 9007-20-9; 9003-01-4; 76050-42-5; 54182-57-9	.06
Triethanolamine	102-71-6	.051
Glycerin	56-81-5	.2
Perfume - CE-60451 - HappyCreations		.15
Tocopheryl Acetate	58-95-7 ; 7695-91-2; 52225-20-4	.01

SAFETY WARNINGS / INSTRUCTIONS

This product is intended for external use only and should not be ingested.
Use of this product by children should be supervised by an adult.

IMPURITIES & TRACE MATERIALS

The below information is a list of impurities & trace materials declared by the manufacturer / responsible person as being present within the product.
If no materials are identified it is understood that no impurities or trace materials have been disclosed to Delphic HSE Solutions at the time of review.

Substances	CAS Number	[Product]

EXPOSURE SCENARIO

Intended Consumer: Children 3+
Single Exposure: 2 g | 5 x Day
Retention Factor: 0.5

Exposure to Neat Product:

Body Site(s): Hands
Surface Area: 437 cm^2
Exposure Level: 11.442 mg/cm^2
Exposure Time: Left on

Minimum Expected Body Weight 14kg
Diluted in use: No
Retained Exposure: 357.143 mg/kg/day

Exposure to Diluted Product: Product Not Diluted in Use

Manufacturers Instructions for Use

No specific instructions for use were provided for review.

Information on Previous Sales / Complaints

Details of previous sales and complaints data has not been supplied for review. This assessment is, therefore, completed on the assumption that no previous health-related complaints have been reported by consumers. If this is incorrect the manufacturer/responsible person must notify Delphic HSE of all such reported complaints so that the assessment can be updated.

Going forward the manufacturer/responsible person must ensure that details of any concerns or complaints relating to consumer safety and adverse health effects are provided to Delphic HSE so that the safety assessment can be updated accordingly.

Information on User Trials / Product Testing

Details of user trials/additional product safety testing have not been supplied for review, and this assessment is conducted on the basis that no such testing has been undertaken. Should this be inaccurate, or additional testing be conducted in the future, Delphic HSE should be notified of the details of such testing so this safety assessment can be updated accordingly.

Product Review

Hand sanitizer intended for use by children aged 3 years plus, juveniles and adults. The hand sanitizer may fall under the regulation of OTC drugs in US, and the manufacturer must ensure that the product complies with relevant requirements. In this review, the product is assessed as a household product as required by the client, and toxicological risk is assessed in this report only.

The formulation contains Triethanolamine, the formation of nitrosamines should be avoided.

The formulation contains 67% Ethyl Alcohol, which is a narcotic solvent. To prevent non-standard and excessive exposure to this ingredient by young children, it would be prudent to label "Use of this product by children should be supervised by an adult" and "This product is intended for external use only and should not be ingested".

The fragrance is present in the formulation at 0.1%. According to the IFRA statement, the fragrance can be used in hand creams up to 20%.

The manufacturer must ensure that all raw materials are of suitably safe US grades.

Overall, the product is considered suitable for its intended use.

Skin Toxicity - Neat Product

Repeated or prolonged skin contact may lead to defatting of the skin which can if prolonged cause dryness irritation and possible cracking
Exposure to this product is unlikely to result in photo-toxic effects.
There are low levels of substances present in this product which are known to cause an allergic reaction. The concentrations are sufficiently low not to present a risk of inducing allergy. However people already sensitised may show an adverse reaction when using this product. The identity of these ingredients will be shown on the label, enabling those potentially affected to avoid contact.
Unlikely to produce systemic toxicity following skin contact.

Eye Toxicity - Neat Product

May irritate the eye.

Oral Toxicity - Neat Product

The product contains ingredients that if swallowed, may cause depression of the central nervous system. Depending on dose, symptoms could include drowsiness, unconsciousness and in extreme cases possibly even death.

Inhalation Toxicity

The product contains volatile liquids which will release a vapour during normal use, which may be absorbed via the lung.
Prolonged inhalation of high concentrations of the aerosol and derived vapours may induce drowsiness, headache, loss of co-ordination and narcosis. In severe over exposure this may lead to coma and death.

CONCLUSIONS & RECOMMENDATIONS

Required Safety Labelling

This product is intended for external use only and should not be ingested.
Use of this product by children should be supervised by an adult.

Overall Safety & Compliance

Considered safe for use as a household product

Review of the documented information for the ingredients in the formulation for this product suggests that it is not expected to be Toxic, Highly Toxic, Corrosive, Irritant or a Strong Sensitiser as defined in 16 CFR 1500.3(b)(5), 1500.3(b)(6), 1500.3(b)(7), 1500.3(b)(8) and 1500.3(b)(9). and is considered to be safe for use as a household product.

Toxicological & Regulatory Assessor

Tian Li

Dr L Tian, BEng, PhD, MRSB, CBiol, DCST, UKRT, ERT 21 Jun 2018

This report consists of 4 pages plus a Regulatory and Ingredient Data Annex. It is only valid as the original, complete document.

Preface to Annexes

Annex II - Ingredient Data

Physical/Chemical and Toxicological data presented within these reviews are representative of publicly available data and provided for informational purposes only. Sources of data are identified (typically in brackets) following each data point, and there may be multiple data points for any given toxicological endpoint.

Margins of Safety (MoS) are calculated where suitable data are available, and may related to mg/kg, µg/cm^2 or percentage-based indications of safety.

MoS based on systemic (mg/kg) effects are calculated as 'Point of Departure (PoD)' / 'Systemic Exposure Dose (SED)', where:

PoD = Data point considered to be indicative of a 'safe' level of exposure. This may be an animal-derived No Observed Adverse Effect Level (NOAEL) or a value indicated as being safe to humans. In the case of the latter this would typically be in the form of an ADI (Acceptable Daily Intake) or DNEL (Derived No Effect Level) established by a governmental or scientific committee / body.

SED = (Product Used (mg) x Retention Factor x Concentration of Material in Product x Dermal Absorption) / intended user body weight (kg)

In the absence of material specific data a dermal absorption of 100% is assumed.
Where an animal-derived NOAEL is used as the PoD an MoS greater than 100 is typically considered acceptable for indicating safety to consumers.
For PoD based on established safe levels in humans an MoS of greater than 1 is typically considered as acceptable for indicating safety to consumers.

MoS based on localised (µg/cm^2) effects are calculated as 'Point of Departure (PoD)' / 'Dermal Exposure', where:

PoD = Data point considered to be indicative of a 'safe' level of exposure. This would typically be a µg/cm^2 value identified from either a Local Lymph Node Assay (LLNA) or Human Repeat Insult Patch Test (HRIPT).

Dermal Exposure = (Product Used (µg) x Retention Factor x Concentration of Material in Product) / Surface Area of Application

MoS based on percentage data are calculated as 'Point of Departure PoD' / 'Ingredient Concentration in Product', where:

PoD = Data point considered to be indicative of a 'safe' level of exposure. Typically a percentage identified as safe for use within a leave-on consumer product, as established by legislation or by a governmental or scientific committee / body.

Ingredient Concentration in Product = Concentration of Material in Finished Product x Retention Factor
(As safe levels are typically identified for leave-on products the retention factor is included within the calculation to account for use in rinse-off products)
For PoD based on established safe levels in finished products an MoS of greater than 1 is typically considered as acceptable for indicating safety to consumers.

Retention Factor is an estimation of the amount of product in prolonged contact with the skin under normal conditions of use, and expressed as the decimal form of a percentage. A retention factor of 1 relates to 100% of the product staying in prolonged contact with the skin and is typically used for all leave-on products. All other products have retention factors as determined by typical conditions of use, and these are presented on page 2 of the assessment under 'Exposure Scenario'.

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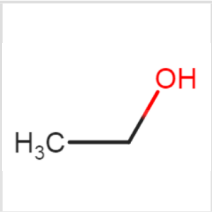
ANNEX I - REGULATORY CONTROLS

Substance: Ethyl Alcohol

CAS: 64-17-5

Function: Antifoaming; Antimicrobial; Astringent; Masking; Solvent; Viscosity controlling

Concentration in Product: 67%



Regulatory Listings

Europe:

EINECS: 200-578-6
EU GHS Classification: H225 Highly flammable liquid and vapour (Flam. Liq. 2) (*harmonised classfication*)

REACh Annex XVII: Not Controlled
REACh SVHC: Not Controlled

USA

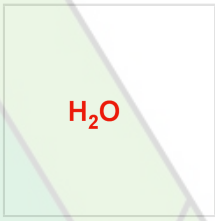
Chemical Inventory: Listed
California Prop 65: Ethanol is listed as a Carcinogen and developmental toxin (listed as in alcoholic beverages)
Cosmetic Regulation: Alcohol
Not Controlled

Substance: Aqua

CAS: 7732-18-5

Function: Solvent

Concentration in Product: 31.735%



Regulatory Listings

Europe:

EINECS: 231-791-2
EU GHS Classification: Unclassified

REACh Annex XVII: Not Controlled
REACh SVHC: Not Controlled

USA

Chemical Inventory: Listed as existing; Water
California Prop 65: Not Listed
Cosmetic Regulation: Water
Not Controlled

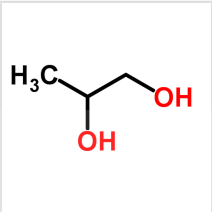
ANNEX I - REGULATORY CONTROLS

Substance: Propylene Glycol

CAS: 57-55-6; 4254-14-2

Function: Humectant; Skin Conditioning; Solvent; Viscosity Controlling

Concentration in Product: 0.5%



Regulatory Listings

Europe:

EINECS: 200-338-0
EU GHS Classification: Not Classified
REACh Annex XVII: Not Controlled
REACh SVHC: Not Controlled

USA

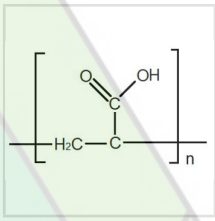
Chemical Inventory: Listed as 57-55-6
California Prop 65: Not listed
Cosmetic Regulation: Propylene Glycol
Not Controlled

Substance: Carbomer (NOS)

CAS: 9007-16-3 ; 9007-20-9; 9003-01-4; 76050-42-5; 54182-57-9

Function: Viscosity Controlling; Emulsion Stabilising; Gel Forming

Concentration in Product: 0.25%



Regulatory Listings

Europe:

EINECS: -
EU GHS Classification: Unclassified
REACh Annex XVII: Not Controlled
REACh SVHC: Not Controlled

USA

Chemical Inventory: Listed
California Prop 65: Not listed
Cosmetic Regulation: Carbomer
Not Controlled

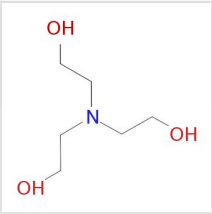
ANNEX I - REGULATORY CONTROLS

Substance: Triethanolamine

CAS: 102-71-6

Function: Surfactant; Buffering; Emulsifying; Masking

Concentration in Product: 0.21%



Regulatory Listings

Europe:

EINECS: 203-049-8
EU GHS Classification: Not classified.

REACh Annex XVII: Not Controlled
REACh SVHC: Not Controlled

USA

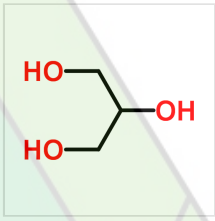
Chemical Inventory: Listed as existing; Ethanol, 2,2',2"-nitrilotris-
California Prop 65: Not Listed
Cosmetic Regulation: Triethanolamine
Not Controlled

Substance: Glycerin

CAS: 56-81-5

Function: Denaturant; Humectant; Hair Conditioning; Oral Care; Perfuming; Skin Protecting; Viscosity Controlling

Concentration in Product: 0.2%



Regulatory Listings

Europe:

EINECS: 200-289-5
EU GHS Classification: Unclassified
REACh Annex XVII: Not Controlled
REACh SVHC: Not Controlled

USA

Chemical Inventory: Listed as existing; 1,2,3-Propanetriol
California Prop 65: Not listed
Cosmetic Regulation: Glycerin
Not Controlled

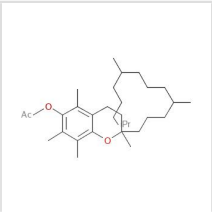
ANNEX I - REGULATORY CONTROLS

Substance: Tocopheryl Acetate

CAS: 58-95-7 ; 7695-91-2; 52225-20-4

Function: Antioxidant; Skin conditioning

Concentration in Product: 0.005%



Regulatory Listings

Europe:

EINECS: 200-405-4; 231-710-0
EU GHS Classification: Unclassified
REACh Annex XVII: Not Controlled
REACh SVHC: Not Controlled

USA

Chemical Inventory: Listed as 7695-91-2
California Prop 65: Not listed
Cosmetic Regulation: Tocopheryl Acetate
Not Controlled

Substance: Perfume - CE-60451 - HappyCreations

CAS:

Function:

Concentration in Product: 0.1%

No Structure Available

Regulatory Listings

Europe:

EINECS: -
EU GHS Classification: H226 Flammable liquid and vapour.
H315 Causes skin irritation.
H317 May cause an allergic skin reaction.
H319 Causes serious eye irritation.
H400 Very toxic to aquatic life.
H410 Very toxic to aquatic life with long lasting effects.
REACh Annex XVII: Not controlled
REACh SVHC: Not controlled

USA

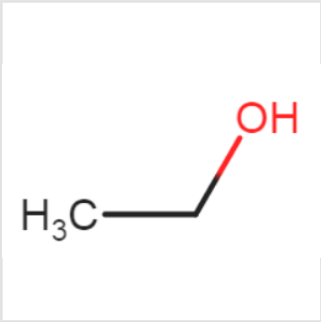
Chemical Inventory:
California Prop 65:
Cosmetic Regulation: Fragrance
Not controlled

ANNEX II - INGREDIENT DATA

Substance: Ethyl Alcohol
CAS: 64-17-5
Function: Antifoaming; Antimicrobial; Astringent; Masking; Solvent; Viscosity controlling

Concentration in Product: 67%
Dermal Exposure Level 7.66590 mg/cm^2
Daily Body Burden 239.28571429mg/kg

Chemical Structure



Physical/Chemical Characteristics

Appearance	Liquid
Boiling Point	78°C
Flammability	Highly
Flash Point	12°C
Melting Point	-114°C
Vapour Pressure	59.3 mm Hg
Water Solubility	1.00E+06 mg/L

Toxicological Summary

The below information is a summary of the toxicological profile for this raw material, including a description of general hazards associated with the material as well as discussion of the most critical studies relating to the overall safety in use of this substance/mixture in consumer products. Details of the available toxicological data can be found overleaf.

Overall Toxicity Review:

A narcotic substance that is readily absorbed via the GI Tract and Lungs, but which is poorly absorbed across the skin (OECD, SIDS). Due to the poor rates of absorption, skin contact is unlikely to result in significant adverse effects and the substance is also not a skin irritant or sensitiser. Moreover, most of the substance rapidly evaporates after application. It possesses moderate potential to irritate the eye.

ECHA has set a dermal DNEL as 206 mg/kg bw/day for the general population.

Overall the use of this material in products where ingestion and inhalation will not be a route of exposure is not expected to pose a significant risk. However misuse (either intentional or accidental) of products containing this substance can lead to severe adverse effects, including Narcosis and Death. As such it is recommended that this material is denatured such as to make it unpalatable, especially in the case of products that children may be exposed to.

Margin(s) of Safety

Maximum Recommended Exposure	206.0000mg/kg	**Exposure from Product**	239.285714mg/kg	**Margin of Safety**	.860895522

ANNEX II - INGREDIENT DATA

Ethyl Alcohol

Acute Toxicity		LD50 = 10470 mg/kg
Acute Oral Toxicity, Lethality [OECD 401, OECD 423, OECD 425]		
Rat	Oral, NOS	
Repeated Dose		0, 5, 10, 20g/kg of 16.25% solution; 21/sex/dose NOAEL = 10 g/kg/day (= 1.625 g/kg/day ethanol)
90-Day Oral Toxicity Study [OECD 408, OECD 409]		
Rat	Oral, Gavage	
Skin Irritation		Not irritating
In vivo skin irritation [Other]		
Rabbit	(Route)	

ANNEX II - INGREDIENT DATA

Substance: Aqua
CAS: 7732-18-5
Function: Solvent

Concentration in Product: 31.735%
Dermal Exposure Level 3.63101 mg/cm^2
Daily Body Burden 113.33928571mg/kg

Chemical Structure

H_2O

Physical/Chemical Characteristics

Boiling Point	100°C
Appearance	Clear colourless
Flammability	Not flammable
Flash Point	not flammable
Molecular Mass	18
Melting Point	0°C
Odour	none
pH	7
Specific Gravity	1

Toxicological Summary

The below information is a summary of the toxicological profile for this raw material, including a description of general hazards associated with the material as well as discussion of the most critical studies relating to the overall safety in use of this substance/mixture in consumer products. Details of the available toxicological data can be found overleaf.

Overall Toxicity Review:

A ubiquitous chemical substance that is the basis for all known forms of life. Use in consumer products is not expected to result in any Acute or Chronic Toxicity following typical exposures.

Margin(s) of Safety

ANNEX II - INGREDIENT DATA

Aqua

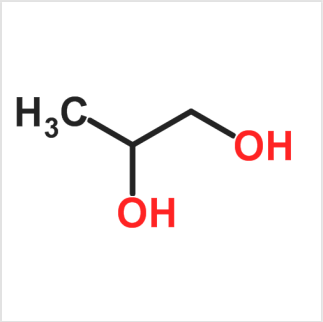
Details on specific toxicological studies related to endpoints of concern are not available for Aqua, please see the previous page for a justification of safety based on history of use &/or weight of evidence.

ANNEX II - INGREDIENT DATA

Substance: Propylene Glycol
CAS: 57-55-6; 4254-14-2
Function: Humectant; Skin Conditioning; Solvent; Viscosity Controlling

Concentration in Product: 0.5%
Dermal Exposure Level 0.05721 mg/cm^2
Daily Body Burden 0.41071429mg/kg

Chemical Structure



Physical/Chemical Characteristics

Melting Point	-60°C
Boiling Point	187.6°C
Log Kow	0.92
Water Solubility	1000000 mg/L
Vapour Pressure	0..129 mm Hg at
Molecular Mass	76.0942
Density	0.785 g/ ml
Appearance	Colourless liquid
Boiling Point	189C

Toxicological Summary

The below information is a summary of the toxicological profile for this raw material, including a description of general hazards associated with the material as well as discussion of the most critical studies relating to the overall safety in use of this substance/mixture in consumer products. Details of the available toxicological data can be found overleaf.

Overall Toxicity Review:

A widely used solvent, which in most cases is of low irritation potential, used in cosmetics and pharmaceuticals. Not allergenic. Approximately 1% of raw material suppliers classified it as skin and eye irritating, whilst the majority have not classified it. Not carcinogenic. Minimal reprotoxicity. Evidence suggests that propylene glycol is more toxic to children than it is to adults and there is a report of ingestion of approximately 75 ml of a 2% concentration of propylene glycol (1.5gm propylene glycol) giving rise to narcosis and acidosis in a 2 year old child (a dose of approximately 125 mg/kg for a 12 kg infant). Overall the use of this material at typical levels would not be expected to pose an undue risk of significant adverse effects. However, caution should be used when considering it's use in Toys due to the increased toxicity to children.

Acceptable Daily Intake (ADI) of 0 - 10 mg/kg bw/day allocated by the Joint FAO/WHO Expert Committee on Food Additives (JECFA) TDI of 5 mg/kg bw/day established by the Scientific Committee on Food (SCF).
JECFA ADI of 25 mg/kg/d was established for propylene glycol in 1973 (evaluated in 2002).

Margin(s) of Safety

Maximum Recommended Exposure	25.0000mg/kg	**Exposure from Product**	0.410714mg/kg	**Margin of Safety**	60.8695652

Physical/Chemical and Toxicological data used in generating these review are representative of publicly available data, and is provided for information purposes only.
For further information on any of the data or conclusions reported here, or for information Margins of Safety and how they are Calculated, please contact Delphic HSE.

ANNEX II - INGREDIENT DATA

Propylene Glycol

Acute Toxicity Acute Toxicity, Lethality [Other] Rabbit [NOS]	Oral, Water	LD50: 18500 mg/kg
Acute Toxicity Acute Toxicity, Lethality [Other] Rabbit [NOS]	Dermal	LD50: 20800 mg/kg
Acute Toxicity Acute Toxicity, Lethality [Other] Mouse [NOS]	Oral, Gavage	LD50 = 22000 mg/kg
Acute Toxicity Acute Toxicity, Lethality [Other] Rabbit [NOS]	Subcutaneous	LDL0 = 6300 mg/kg
Acute Toxicity Acute Toxicity, Lethality [Other] Human	Oral, Feed	TDLo: 79gm/kg/56W-I
Acute Toxicity Acute Toxicity, Lethality [Other] Dog [NOS]	Oral, Feed	LD50: 22000 mg/kg
Acute Toxicity Acute Toxicity, Lethality [Other] Rat [NOS]	Oral, Feed	LD50: 22000 mg/kg
ADME (Test) (Species)	(Route)	Propylene glycol is oxidized to pyruvic acid, acetic acid, lactic acid (which then enters the metabolic pool) and propionaldehyde. The substance is mainly eliminated in urine, but renal clearance decreases with dose
ADME (Test) (Species)	(Route)	Relative dermal absorption of the applied dose was estimated to be 23% (0.96%/h)
Carcinogenicity Carcinogenicity studies [Other] Rat [NOS]	Oral, Feed	NOAEL: 1700 mg/kg bw/day (Male); 2100 mg/kg bw/day (Female) Duration: 2 years
Carcinogenicity Carcinogenicity studies [Other] Dog [NOS]	Oral, Feed	Dose: 0, 2000 mg/kg bw/d, 5000 mg/kg bw/d Duration: 104 weeks Result: No increase in tumour incidence
Eye Irritation In vivo Eye Irritation [Other] Rabbit [NOS]	Instillation	cornea opacity score 0/4; iris score 0.1/2; conjunctivae score 0.4/3; chemosis score 0/4
Genotoxicity Bacterial reverse mutation test (Ames) [OECD 471] Bacteria [NOS]	In vitro exposure	Negative Ames Test +/- S9 activation
Genotoxicity Mammalian chromosome aberration test [OECD 473] Bacteria [NOS]	In vitro exposure	Negative chromosomal aberration (human lymphocytes) +/- S9 activation
Repeated Dose Repeat Dose Oral Toxicity Study [Other] Rat [NOS]	Oral, Feed	NOAEL rat - 1700 mg/kg bw/day (in feed); female rat - 2100 mg/kg bw/day
Repeated Dose Repeat Dose Oral Toxicity Study [Other] (Species)	Oral, Feed	NOAEL Oral - Cat: 443 mg/kg bw/day
Reproductive Toxicity In vivo reproductive toxicity study [Other] Mouse [NOS]	Oral, Feed	NOAEL for toxicity/ fertility/ developmental effects:10100 mg/kg bw/day
Skin Irritation Draize Test [OECD 404] Rabbit [NOS]	Dermal	Non irritating
Skin Sensitisation Maximisation Test [Other] Guinea Pig [NOS]	Dermal	70% propylene glycol with water. tested 20 subjects. Not a sensitiser.
Skin Sensitisation Local Lymph Node Assay [OECD 429, OECD 442A, OECD 442B] Mouse [NOS]	Dermal	Dose: 50% and 100% Result: For 50% solution, 1.2; For 100% test substance, 1.6

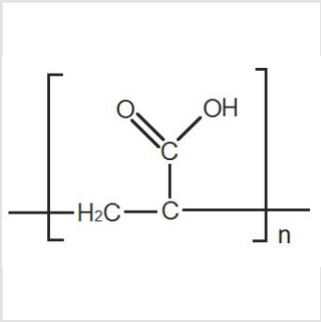
Physical/Chemical and Toxicological data used in generating these review are representative of publicly available data, and is provided for information purposes only.
For further information on any of the data or conclusions reported here, or for information Margins of Safety and how they are Calculated, please contact Delphic HSE.

ANNEX II - INGREDIENT DATA

Substance: Carbomer (NOS)
CAS: 9007-16-3 ; 9007-20-9; 9003-01-4; 76050-42-5; 54182-57-9
Function: Viscosity Controlling; Emulsion Stabilising; Gel Forming

Concentration in Product: 0.25%
Dermal Exposure Level 0.02860 mg/cm^2
Daily Body Burden 0.89285714mg/kg

Chemical Structure



Physical/Chemical Characteristics

Appearance	White Powder to
Flammability	Autoignition @
Odour	Acetic (powder)
pH	2.5 to 3 @ 1%
Specific Gravity	1.4 @20C
Viscosity	>11,000 cP
Water Solubility	Will swell in water

Toxicological Summary

The below information is a summary of the toxicological profile for this raw material, including a description of general hazards associated with the material as well as discussion of the most critical studies relating to the overall safety in use of this substance/mixture in consumer products. Details of the available toxicological data can be found overleaf.

Overall Toxicity Review:

A polymeric thickening agent with low potential to cause irritation or allergy. The grade used must not contain benzene in order to comply with the EU Cosmetic regulations and Toy Safety Regulations. A solution of 0.5% Carbopol has a viscosity at room temperature of 50,000 cP , similar to tomato ketchup (water has a viscosity of 1 cP at room temperature). Typically used at 0.25 to 1% the carbopol will give an acid pH of c.3 due to acid side groups. The solution should be neutralised with alkali if used in skin contact or applications that are pH crtitical.

A NOAEL was not available for review however, under normal conditions of use, incorporation at low levels within a consumer product would not be expected to pose an undue risk of significant adverse effects.

Margin(s) of Safety

ANNEX II - INGREDIENT DATA

Carbomer (NOS)

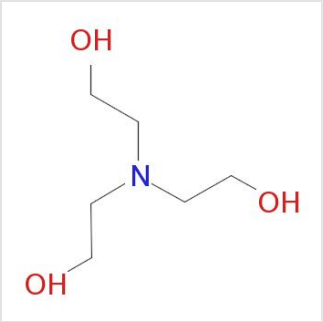
Acute Toxicity Acute Toxicity, Lethality [Other] Rat	Oral, NOS	LD50 > 2500 mg/kg
Acute Toxicity Acute Toxicity, Lethality [Other] Rabbit	Dermal	LD50 > 3000 mg/kg
Repeated Dose Repeat Dose Oral Toxicity Study [Other] Rat	Oral, Feed	No significant effects in dogs fed with resin as 5% of diet for 6.5 months.
Repeated Dose Repeat Dose Oral Toxicity Study [Other] Dog	(Route)	No significant effects in rats fed with resin as 5% of diet for 6.5 months.
Skin Sensitisation Repeat Insult Patch Test (RIPT) [Other] Human	Dermal	200 subjects had Carbomer 934 (100% neat resin) applied to the skin daily for 5 days under occlusive patches. It was reapplied for 48hrs following a 3-week rest period. No evidence of skin sensitisation or irritation.

ANNEX II - INGREDIENT DATA

Substance: Triethanolamine
CAS: 102-71-6
Function: Surfactant; Buffering; Emulsifying; Masking

Concentration in Product: 0.21%
Dermal Exposure Level 0.02403 mg/cm^2
Daily Body Burden 0.07050000mg/kg

Chemical Structure



Physical/Chemical Characteristics

Appearance	Water white liquid
Boiling Point	190 - 193 °C at 7
Evaporation Rate	<0.01
Flash Point	190.5C
Molecular Mass	149
Melting Point	18 - 21 C
Odour	Mild ammoniacal
pH	10.5 - 11.5 at 149
Specific Gravity	1.26 @ 20C

Toxicological Summary

The below information is a summary of the toxicological profile for this raw material, including a description of general hazards associated with the material as well as discussion of the most critical studies relating to the overall safety in use of this substance/mixture in consumer products. Details of the available toxicological data can be found overleaf.

Overall Toxicity Review:

A base soluble in water and organic solvents. In free base form can be irritant to skin and eye, when ionised has lower potential to cause irritation. Generally used to neutralise polymers in a formulation. Has high Oral and Dermal LD50s and little allergenic capacity.

In a subchronic dermal toxicity in rats, a NOAEL of 125 mg/kg/day was determined: at the LOAEL of 250 mg/kg/day, relative weight of kidneys was increased. (ECHA registration dossier). ECHA has set a dermal DNEL as 3.1 mg/kg bw/day for the general population.

Care should be taken to avoid the formation of nitrosamines within a product formulation.

CIR report indicates that Triethanolamine is used in leave on products in the range of 0.0002-6% and in rinse off products in the range of 0.0003-19%.

Margin(s) of Safety

Maximum Recommended Exposure	2.5000%	**Exposure from Product**	0.210000%	**Margin of Exposure**	11.90
Maximum Recommended Exposure	125.0000mg/kg	**Exposure from Product**	0.070500mg/kg	**Margin of Safety**	1773.04965
Maximum Recommended Exposure	3.1000mg/kg	**Exposure from Product**	0.070500mg/kg	**Margin of Safety**	43.9716312

ANNEX II - INGREDIENT DATA

Triethanolamine

Acute Toxicity Acute Oral Toxicity, Lethality [OECD 401, OECD 423, OECD 425] Rat	Oral, Gavage	LD50 = 6400 mg/kg bw
Acute Toxicity Acute Inhalation Toxicity, Lethality [OECD 403, OECD 436] Rat	Inhalation	LC50: 1.8mg/m3 (saturated TEA atmosphere)
Acute Toxicity Acute Dermal Toxicity, Lethality [OECD 402] Rabbit	Dermal	LD50: >2000mg/kg bw
ADME In vitro skin absorption [OECD 428] Human	In vitro exposure	In vitro, the absorption of TEA through human skin was low under conditions simulating perceived cosmetic use, i.e., 1-5% at pH 7.0; approximately 5.5 - 9.4% of the dose was recovered in the skin after 24 or 72 h, however, only 0.28 - 0.43% was recovered in the receptor fluid.
Carcinogenicity Carcinogenicity studies [OECD 451] Mouse	Dermal	Mouse, dermal exposure, 104 (males) or 105 (females) weeks. Dose for females: 100, 300, 1000 mg/kg bw/day; Dose for males: 200, 630, 2000 mg/kg bw/day. NOAEL <= 100 mg/kg bw/day for both males and females [OECD 451] (ECHA 2004); Classified as group 3 Not classifiable as to its carcinogenicity to humans by IARC (2000)
Eye Irritation Draize, Standard [OECD 405] Rabbit	Instillation	Not irritating to rabbit eyes. Maximum mean total score (MMTS) at 0.01 was 0, 0.03 was 1, and 0.1 mL was 4. Not considered irritating.
Genotoxicity Bacterial reverse mutation test (Ames) [OECD 471] Bacteria	In vitro exposure	Negative in Ames test in TA1535, TA1537, TA97, TA98 and TA100 with and without metabolic activation.
Repeated Dose 90-Day Oral Toxicity Study [OECD 408, OECD 409] Rat	Oral, Feed	91 days, dose: 0; 250; 500; 1000 mg/kg bw, oral NOAEL (rat): 1000 mg/kg/day [OECD 408] (ECHA 1989) Prolonged repeated ingestion in laboratory animals has adverse effects on the Kidney ((MSDS, Huntsman).
Repeated Dose 90-day Dermal Toxicity Study [OECD 411] Rat	Dermal	Dermal NOAEL for systemic effects: 125 mg/kg bw/day (Male) or 500 mg/kg bw/day (Female) based on increased absolute and relative kidney weight. Doses tested: 125; 250; 500; 1000; 2000 mg/kg bw/day
Reproductive Toxicity Reproduction/Developmental Toxicity Screening Test [OECD 421] Rat	Oral, Gavage	P0 NOAEL > 1000 mg/kg bw/day for systemic toxicity and reproductive performance and fertility; F1 NOAEL = 300 mg/kg bw/day for developmental toxicity. Doses tested 100, 300, 1000 mg/kg bw/day.
Skin Irritation Draize Test [OECD 404] Rabbit	Dermal	Not irritating to rabbit skin. Mean erythema score: 0, mean oedema score: 0 at 4, 24, 48 and 72 hours.
Skin Sensitisation Buehler [OECD 406] Guinea Pig	Dermal	Not sensitising in guinea pig maximisation test. Intradermal induction: 2 %, dermal induction: unchanged, dermal challenge: 10%. TEA has a low potential to induce skin sensitisation and does not meet the criteria for classification according to ECHA - CoRAP.

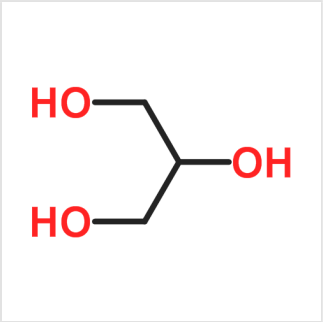
Physical/Chemical and Toxicological data used in generating these review are representative of publicly available data, and is provided for information purposes only.
For further information on any of the data or conclusions reported here, or for information Margins of Safety and how they are Calculated, please contact Delphic HSE.

ANNEX II - INGREDIENT DATA

Substance: Glycerin
CAS: 56-81-5
Function: Denaturant; Humectant; Hair Conditioning; Oral Care; Perfuming; Skin Protecting;

Concentration in Product: 0.2%
Dermal Exposure Level 0.02288 mg/cm^2
Daily Body Burden 0.71428571mg/kg

Chemical Structure



Physical/Chemical Characteristics

Appearance	Colourless syrup
Boiling Point	290˚C
Explosive	Non-explosive
Flammability	400C
Flash Point	177˚C (Open
Log Kow	1.76
Molecular Mass	92.11
Melting Point	18.2˚C
Odour	Sweet

Toxicological Summary

The below information is a summary of the toxicological profile for this raw material, including a description of general hazards associated with the material as well as discussion of the most critical studies relating to the overall safety in use of this substance/mixture in consumer products. Details of the available toxicological data can be found overleaf.

Overall Toxicity Review:

A substance with minimal irritancy properties, which is non-sensitising in Human Patch Tests and negative in the Ames Test. When given orally at 20% in the diet over 2 years showed no adverse effects in rats. Has a high Oral LD50 (greater than 10g/kg in Rats), constitutes around 10% of the Fat found in a typical Human Diet and is readily metabolised on ingestion. Nevertheless ingestion of large amounts of this material can cause an osmotic effect in the GI tract leading to dehydration, nausea and headaches. In humans the LDLo (oral) for these effects has been found to be around 1.428 g/kg bw. Glycerin has GRAS status in the US. (FDA, 21CFR182.1320)

Given the low toxicity profile of this substance, and the fact it is a constituent of a typical human diet, its use in Consumer Products is not expected to produce significant localised or systemic toxicity. CIR report indicates that glycerin is used in leave on products up to 79.2% and baby products in the range of 0.23-21%.

Margin(s) of Safety

Maximum Recommended Exposure	8000.0000mg/kg	**Exposure from Product**	0.714286mg/kg	**Margin of Safety**	11200

ANNEX II - INGREDIENT DATA

Glycerin

Acute Toxicity Acute Toxicity, Lethality [Other] Rat	Oral, Gavage	The acute oral LD50 of glycerine was examined. Results for natural and synthetic glycerine were comparable with an oral LD50 of 27,200 mg/kg.
Acute Toxicity Acute Toxicity, Lethality [Other] Guinea Pig	Oral, NOS	Guinea Pig LD50 oral: 7750mg/kg
Acute Toxicity Acute Toxicity, Non-Lethal [Other] Human	Oral, NOS	TDLo oral 1428mg/kg BEHAVIORAL: HEADACHE GASTROINTESTINAL: NAUSEA OR VOMITING
Acute Toxicity Acute Toxicity, Lethality [Other] Rabbit	Dermal	Rabbit LD50 skin > 10gm/kg
Carcinogenicity Carcinogenicity studies [Other] Rat	Oral, Feed	The carcinogenic potential of glycerin was examined by administrating the test material in the diet for up to two years to rats. Administration of glycerin for up to two years in the diet did not result in an increase in tumor formation.
Eye Irritation In vivo Eye Irritation [Other] Rabbit	Instillation	Glycerin was considered to be nonirritating in 19 laboratories and of questionable irritation in one laboratory.
Genotoxicity Bacterial reverse mutation test (Ames) [OECD 471] Bacteria	In vitro exposure	The mutagencity potential of glycerin was examined using the Ames test. The test material was negative as tested in the Ames test.
Repeated Dose Chronic Toxicity Studies in Rodents [OECD 452] Rat	Oral, Feed	The NOAEL was 8000-10,000 mg/kg bw based on the absence of treatment related effects in high dose animals.
Repeated Dose 90-day Inhalation Toxicity Study [OECD 413] Rat	Inhalation	The NOAEL was 167 mg/m3 based on local irritant effects on the upper respiratory tract.
Repeated Dose Repeat Dose Dermal Toxicity Study [Other] Rabbit	Dermal	There were no effects noted in rabbits dosed 8 hours/day, 5 days/week for 45 weeks with dose levels as high as 4.0 ml/kg.
Reproductive Toxicity Two-Generation Reproduction Toxicity [OECD 416] Rat	Oral, Gavage	Glycerin was administered by oral gavage to groups of male and female rats through two generations. There was no effect noted on growth, fertility and reproductive performance through two generations at a dose level of ~2000 mg/kg/day.
Reproductive Toxicity Prenatal Development Toxicity Study [OECD 414] Rat	Oral, Gavage	A developmental toxicity study was conducted in rats. There was no effect on developmental toxicity of offspring of female rats dosed with glycerin at doses as high as 1310 mg/kg/day.
Skin Irritation Draize Test [OECD 404] Rat	Dermal	Glycerin was considered to be non irritating to the skin in rabbit irritation studies in 14 testing laboratories.
Skin Irritation In vivo skin irritation [Other] Human	Dermal	The dermal irritation potential was examined in 33 humans, 30 female and 3 male. Under the conditions of the study, Glycerine USP (25% concentration) exhibited no clinical irritation when tested in humans.
Skin Sensitisation Repeat Insult Patch Test (RIPT) [Other] Human	Dermal	In a study of 420 patients with eczema, 419 showed no irritation or sensitization when tested with a 50% solution in water. One individual reportedly was sensitized but the study design does not prove that.

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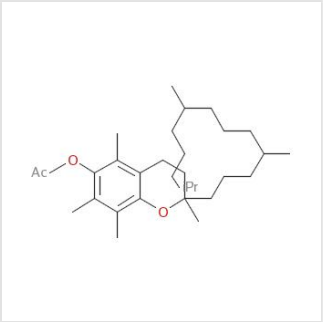
ANNEX II - INGREDIENT DATA

Substance: Tocopheryl Acetate
CAS: 58-95-7 ; 7695-91-2; 52225-20-4
Function: Antioxidant; Skin conditioning

Concentration in Product: 0.005%
Dermal Exposure Level 0.00057 mg/cm^2
Daily Body Burden 0.00075000mg/kg

Chemical Structure



Physical/Chemical Characteristics

Appearance	Viscous slightly
Boiling Point	443°C ; 300°C
Flammability	Ignition 320°C
Flash Point	243°C ; 210°C
Log Kow	12.2
Molecular Mass	472.7
Melting Point	-27.5°C
Specific Gravity	0.96
Viscosity	6589 cP @ 20°C

Toxicological Summary

The below information is a summary of the toxicological profile for this raw material, including a description of general hazards associated with the material as well as discussion of the most critical studies relating to the overall safety in use of this substance/mixture in consumer products. Details of the available toxicological data can be found overleaf.

Overall Toxicity Review:

Vitamin E Acetate is a commonly used as Antioxidant in consumer products.

Available data indicates little risk of localised toxicity; no evidence of skin or eye irritation in experimental animals (1989). Repeat Insult Patch Testing using Human volunteers (HRIPT, 1975) indicated no significant allergenic potential.

As a lipophilic Vitamin (cf. retinoids) there is a potential risk of toxicity from chronic exposure due to accumulation. However, NOAEL values in rodents (e.g. Oral 90 days Rat NOAEL of 500 mg/kg bw/day, 1986) suggest limited capacity for significant toxicological effects following prolonged exposure to the levels typically seen in consumer products.

There was no evidence of genotoxic, carcinogenic or toxicity to reproduction in the available studies.

Both the Scientific Committee on Cosmetic Products and Non-Food Products (SCCNFP) and the Cosmetic Ingredient Review (CIR) Panel have previously released opinions on this material. The SCCNFP concluded that Tocopheryl Acetate did not pose a Hazard to humans and as such did not consider it necessary to impose any restrictions on its use in Leave-on items. The CIR panel concluded that its use at the reported levels of use (up to 36% in leave-on products).

Overall, this material is not expected to produce significant localised or systemic toxicity at typical levels found within consumer products.

Margin(s) of Safety

Maximum Recommended Exposure	36.0000%	**Exposure from Product**	0.005000%	**Margin of Exposure**	7200.00
Maximum Recommended Exposure	500.0000mg/kg	**Exposure from Product**	0.000750mg/kg	**Margin of Safety**	666666.667

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ANNEX II - INGREDIENT DATA

Tocopheryl Acetate

Acute Toxicity Acute Oral Toxicity, Lethality [OECD 401, OECD 423, OECD 425] Rat	Oral, Gavage	LD50 > 10g/kg
Acute Toxicity Acute Dermal Toxicity, Lethality [OECD 402] Rat	Dermal	LD50 > 3g/kg
ADME In vitro skin absorption [OECD 428] Pig	In vitro exposure	Dermal Absorption = 4.2% (5% in AHA Cream, over 18hrs)
Carcinogenicity Carcinogenicity studies [Other] Rat	Oral, Feed	NOAEL > 2g/kg (highest tested dose) (Study conducted to the general principles of OECD 453, however it pre-dates the introduction of the OECD method and no stability of the material in food was done.)
Eye Irritation Draize, Standard [OECD 405] Rabbit	Instillation	Non-irritating
Genotoxicity Mammalian erythrocyte micronucleus test [OECD 474] Mouse	Oral, Feed	Non-mutagenic at up to 200mg/kg/day (highest tested dose)
Genotoxicity Bacterial reverse mutation test (Ames) [OECD 471] Bacteria	In vitro exposure	Negative under Ames Assay conditions up to 5000µg/plate. (S. Typhimurium TA 1535, 97, 98, 100, 102. With and without metabolic activation (Rat S9 Fraction).)
Repeated Dose 90-Day Oral Toxicity Study [OECD 408, OECD 409] Rat	Oral, Gavage	NOAEL = 500mg/kg/day
Repeated Dose 28-day Oral Toxicity Study [OECD 407] Rat	Oral, Gavage	NOAEL > 2g/kg (highest tested dose)
Reproductive Toxicity In vivo reproductive toxicity study [Other] Rat	Oral, Gavage	NOAEL > 1.6g/kg/day (Pre-GLP Study conducted as per the general principles of OECD 414).
Skin Irritation Draize Test [OECD 404] Rabbit	Dermal	Non-irritating
Skin Sensitisation Repeat Insult Patch Test (RIPT) [Other] Human	Dermal	Non-sensitising (203 human volunteers. Induction with 100% Tocopheryl Acetate, 10 times over two weeks. 2-week rest period followed by 3-days of challenge with 100%.)

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ANNEX II - INGREDIENT DATA

Substance: Perfume - CE-60451 - HappyCreations
CAS:
Function:

Concentration in Product: 0.1%
Dermal Exposure Level 0.01144 mg/cm^2
Daily Body Burden 0.35714286mg/kg

Chemical Structure

No Structure Available

Physical/Chemical Characteristics

Toxicological Summary

The below information is a summary of the toxicological profile for this raw material, including a description of general hazards associated with the material as well as discussion of the most critical studies relating to the overall safety in use of this substance/mixture in consumer products. Details of the available toxicological data can be found overleaf.

Overall Toxicity Review:

The supplied fragrance is classified as H317 according to manufacturer's MSDS. When used undiluted in a leave-on cosmetic product at a concentration of up to 1.07% the concentration of each potential allergen will be at least 10 x lower than the concentration shown not to cause allergy in human trials.

It should be noted that the full composition of the fragrance has not been disclosed and therefore the manufacture must ensure that the fragrance does not contain any materials which are prohibited for the intended use.

While the fragrance material at or below a concentration of 1.07% is not expected to result in adverse effects in the majority of users, it must be noted that individuals with a pre-existing allergy to one or more of the components may react adversely at levels lower than this. IFRA guidelines indicate that the fragrance is suitable for use in hand creams at a concentration of 20%.

Margin(s) of Safety

ANNEX II - INGREDIENT DATA

Perfume - CE-60451 - HappyCreations

Details on specific toxicological studies related to endpoints of concern are not available for Perfume - CE-60451 - HappyCreations, please see the previous page for a justification of safety based on history of use &/or weight of evidence.